Expectation, Higher Moments

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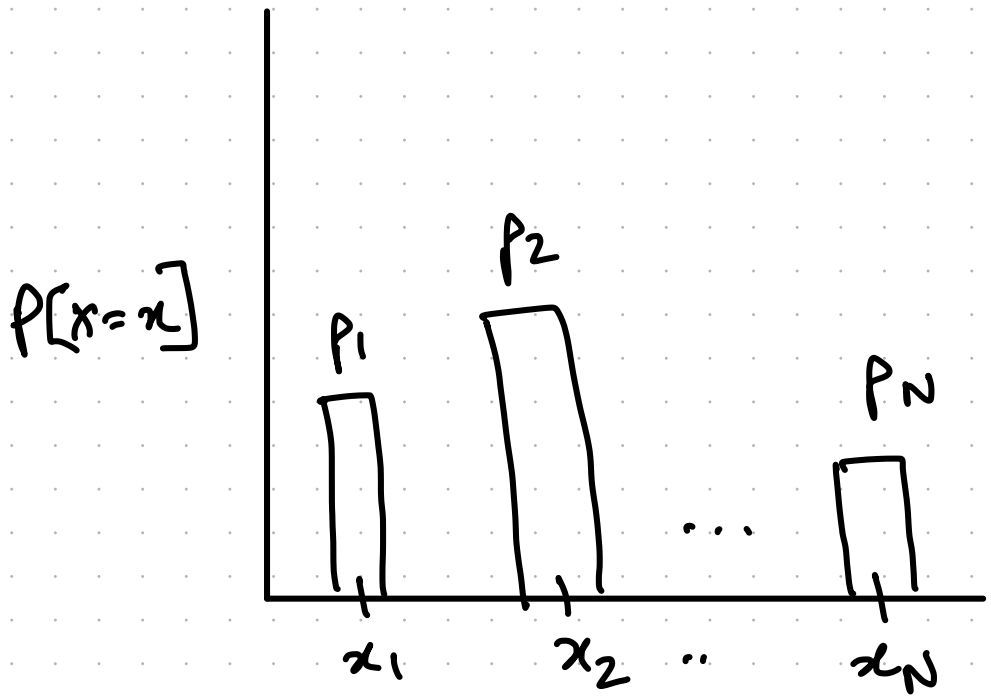
NO!

In expectation, you will lose 50 paise

$$E[X] = \sum_{x \in X(\Omega)} x \, p_X(x)$$

where

$$p_X(x) = P[X = x]$$



$$E[x] = p_1 x_1 + p_2 x_2 + \cdots p_N x_N$$

Q) Coin Toss

- 2 coin toss
- X is r.v. # H.

ω	$X(\omega)$
TT	0
TH	1
HT	1
HH	2

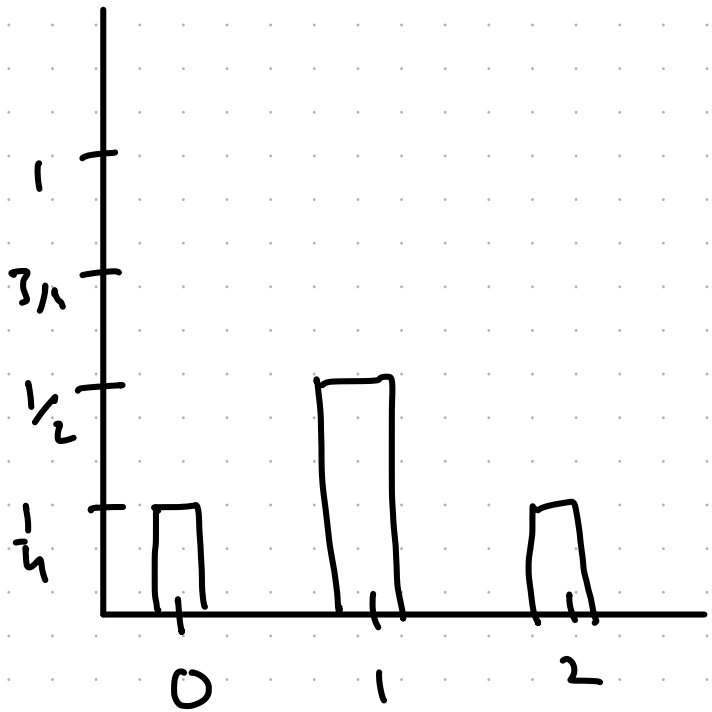
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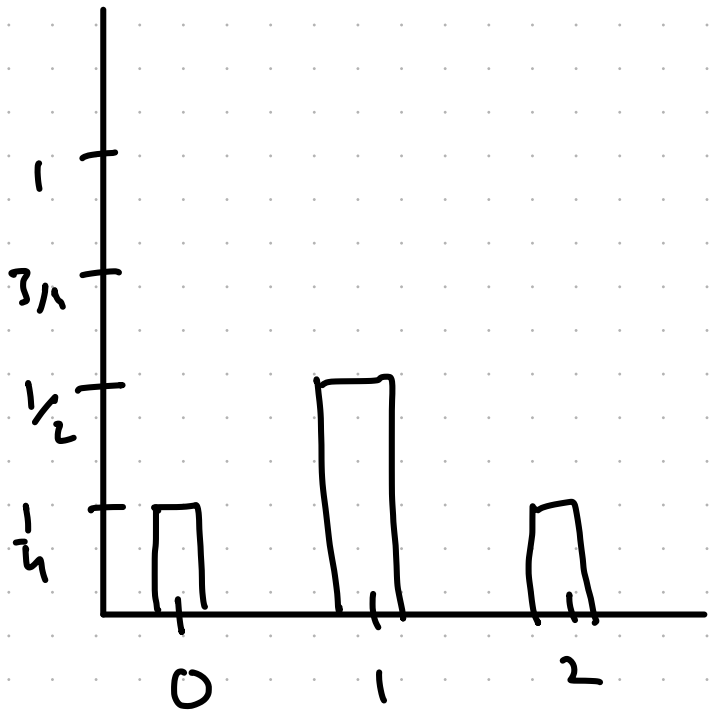


Q) Coin Toss

- 2 coin toss
- X is r.v. # H.

ω	$X(\omega)$
TT	0
TH	1
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HH	2

$E[X] = ?$



$$E[X] = 0 \times \frac{1}{4} + 1 \times \frac{1}{2} + 2 \times \frac{1}{4}$$

$$= 1$$

Q) $X \sim \text{Bernoulli}(p)$

$E[X] = ?$

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$$E[X] = \sum_{x \in \{0,1\}} x \, p_X(x) = 0*(1-p) + 1*p$$

$$= p$$

6) $X \sim \text{Binomial}(n, p)$

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HINT: what is $\sum_{k=0}^{n} p_X(k)$?

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HINT: what is $\sum_{k=0}^{n} p_X(k)$? 1 (as PMF)

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$$E[X] = \sum_{k=0}^{n} k * \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \sum_{k=1}^{n} k * \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

[As $0 \times \ldots = 0$ so start from $k=1$]

$$= n \sum_{k=1} \binom{n-1}{k-1} p^k (1-p)^{n-k}$$

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$$= np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k}$$

$$= np \sum_{m=0}^{n-1} \binom{n-1}{m} p^m (1-p)^{n-1-m}$$

(6) $X \sim \text{Binomial}(n, p)$

$E[X] = ?$

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = np \sum_{m=0}^{n-1} \binom{n-1}{m} p^m (1-p)^{n-1-m}$$

$$= np \sum p_Y(m) \quad \text{where } Y \sim \text{Binomial}(n-1, m)$$

$$= np * 1$$

$$= np$$

Q) $X \sim \text{Poisson}(\lambda)$

$E[X] = ?$

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$$= \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1} e^{-\lambda}}{(k-1)!} = \lambda \sum_{m=0}^{\infty} \frac{\lambda^m \cdot e^{-\lambda}}{m!}$$

$$= \lambda \cdot 1 = \lambda$$

Properties of Expectation

1) Function of X

For any function g:

$$E[g(X)] = \sum_x g(x) p_X(x)$$

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Q). You toss two coins with p = .5

0 heads = 0 Rupees

1 heads = 10 Rupees

2 heads = 25 Rupees

game playing = 20 Rupees

On average what is your return?

Properties of Expectation

1) Function of X

For any function g;

$$E[g(X)] = \sum_x g(x) p_X(x)$$

Q). You toss two coins:

0 heads = 0 Rupees

1 heads = 10 Rupees

2 heads = 25 Rupees

Game playing = 20 Rupees

On average what is your return?

$X \sim \text{Binomial}(2, 0.5)$

$P[X=0] = p_X(0) = \binom{2}{0} (.5)^0 (.5)^{2-0}$

$= 1/4$

$p_X(1) = 1/2$

$p_X(2) = \frac{1}{4}$

Properties of Expectation

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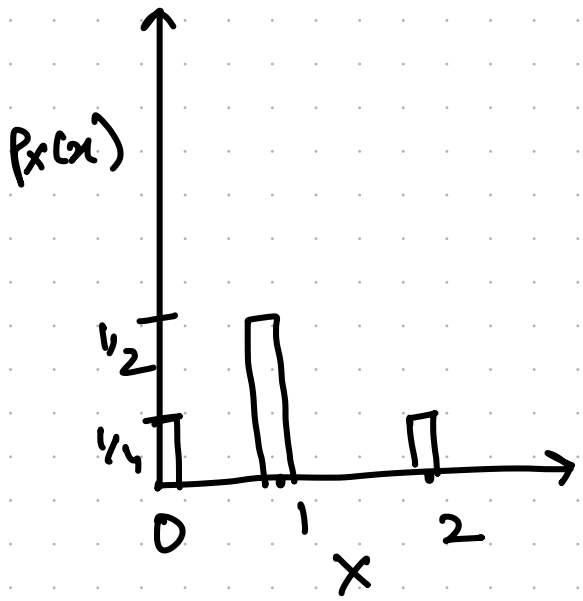
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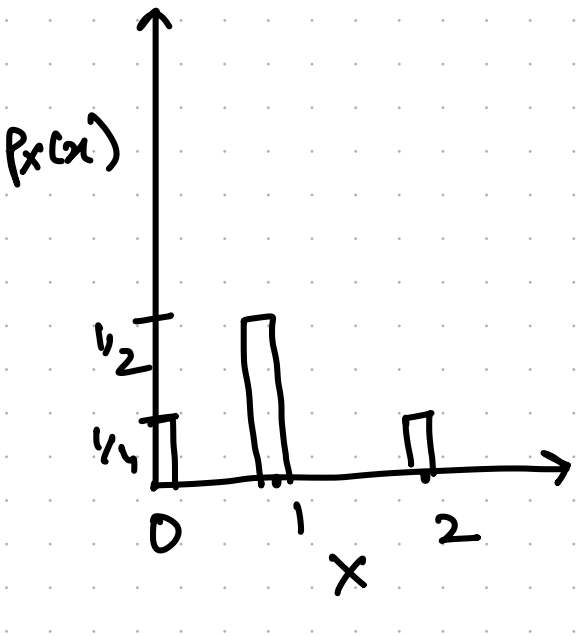
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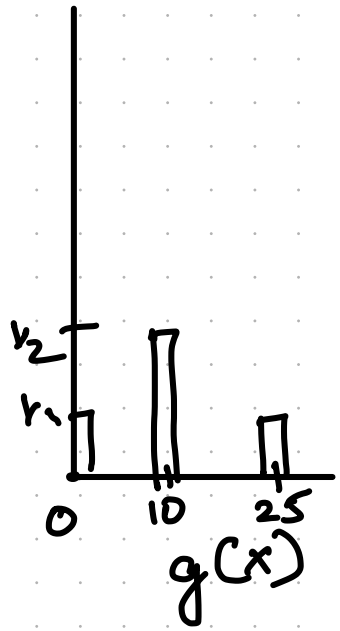
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Properties of Expectation

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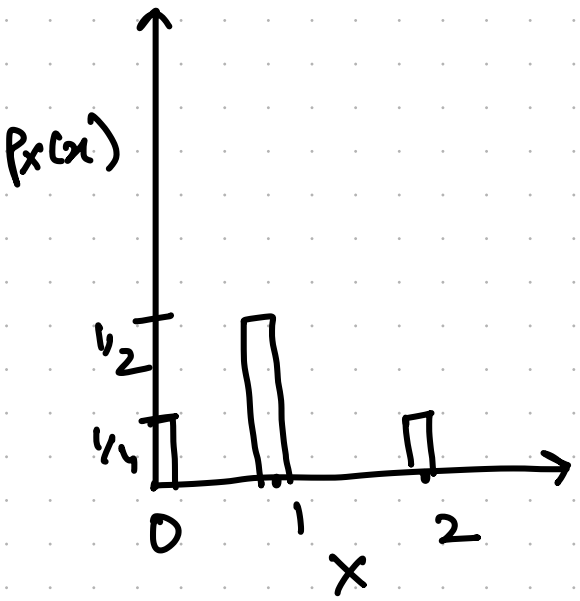
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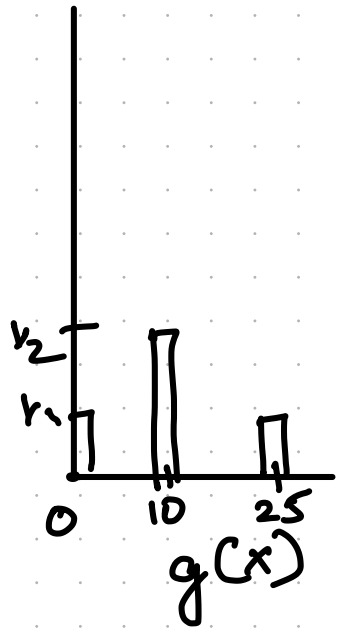
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$$E[g(X)] = 0 \times \frac{1}{4} + 10 \times \frac{1}{2} + 25 \times \frac{1}{4}$$

$$= 5 + 6.25 = 11.25$$

Properties of Expectation

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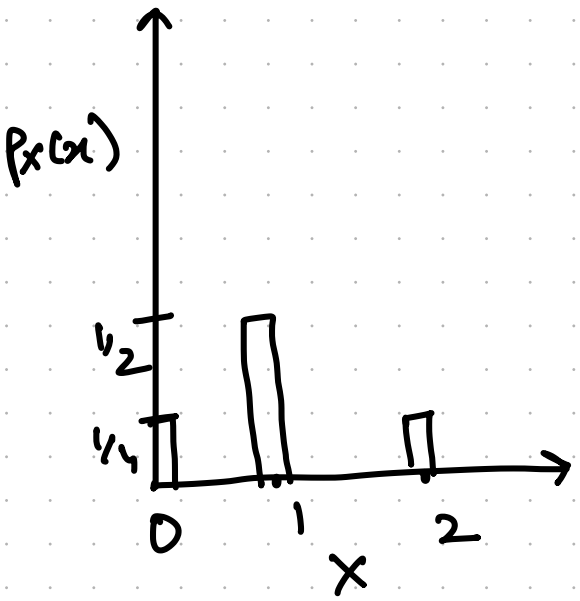
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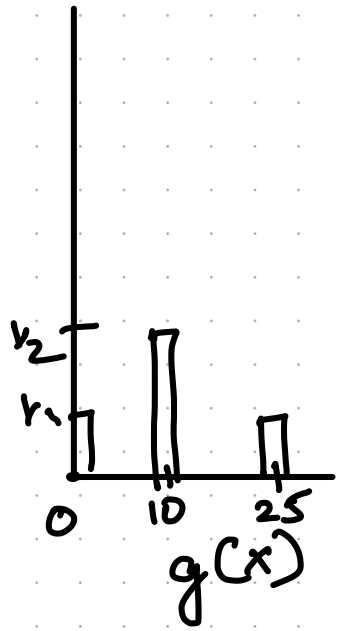
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$E[g(X)] = 11.25$

Expected gain = $E[g(x)] - 20 = -8.75$

Properties of Expectation

1) Scale $E[cX] = c\,E[X]$

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$$E[cX] = \sum_x cx\, p_X(x)$$

$$= c \sum_x x\, p_X(x)$$

$$= c\, E[X]$$

Properties of Expectation

1) Scale $E[cX] = c\,E[X]$

Q) Assume $X \sim \text{Poisson}(\lambda = 5)$ = # orders you sell per day

For each sale you get 1000 INR:

what is Expected money earned daily

Properties of Expectation

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Properties of Expectation

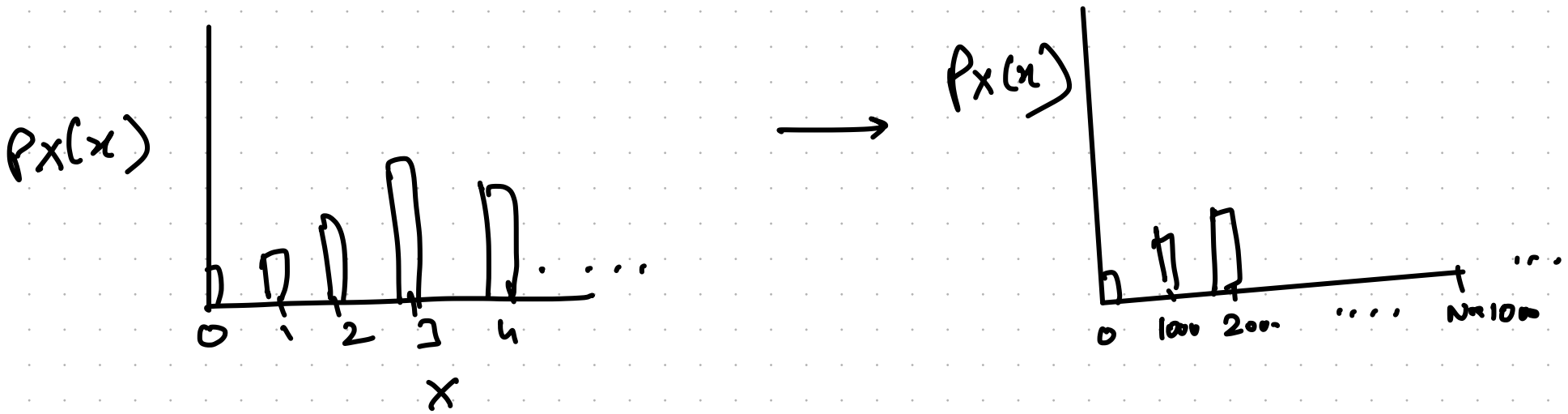
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$$E[g(X)+h(X)] = E[g(X)] + E[h(X)]$$

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$$= \sum_x (g(x)+h(x))\, p_x(x)$$

$$= \sum_x g(x)\, p_x(x) + \sum_x h(x) p_x(x)$$

$$= E[g(x)] + E[h(x)]$$

Properties of Expectation

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$$E[g(X)+h(X)] = E[g(X)] + E[h(X)]$$

Q). $X \sim \text{Poisson}(3)$ denote r.v. for # daily sales.

$g(X) = 1000X$ (Commission per sale)

$$h(X) = \begin{cases} -2000 & ; 0 \text{ sale} \\ -500 & ; 1 \text{ sale} \\ 0 & ; \geq 2 \text{ sales} \end{cases} \quad \left.\vphantom{\begin{cases} -2000 & ; 0 \text{ sale} \\ -500 & ; 1 \text{ sale} \end{cases}}\right\} \text{[Penalty]}$$

what is expected amt. you take home

Properties of Expectation

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Properties of Expectation

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$$E[g(X)+h(X)] = E[g(X)] + E[h(X)]$$

$$= 1000 * 3 + (-2000) * \frac{e^{-3} 3^0}{0!} + (-500) * \frac{e^{-3} 3^1}{1!}$$

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$$= 3000 + (-99) + (-75) \simeq 3000$$

Properties of Expectation

IV) DC shift

$$E[X+c] = E[X]+c$$

-

Properties of Expectation

IV) DC shift

$$E[X+c] = E[X] + c$$

$$= \sum_x (x+c)\, p_X(x)$$

$$= \sum_x x\, p_X(x) + c \sum_x p_X(x)$$

$$= E[X] + c \cdot 1$$

$$= E[X] + c$$

Q) You have option to play 2 games

G_1 : Heads +2000 Rs
Tails −2000 Rs

G_2 : Heads +200 Rs
Tails −200 Rs

What is expected return for both games?

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BUT G_1 is riskier : Larger Variance!

Moments

k^{th} moment of r.v. X is

$$E[X^k] = \sum_x x^k p_X(x)$$

Variance

variance of r.v. showing spread is

$$VAR[X] = E[(X - \mu_X)^2]$$

Moments

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Variance of r.v. showing spread is

$$VAR[X] = E\left[(X - E[X])^2\right]$$

$$= E\left[X^2 - 2X E[X] + E[X]^2\right]$$

$$= E[X^2] - 2E[X] \cdot E[X] + E[X]^2$$

$$= E[X^2] - E[X]^2$$

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X is r.v.
$E[X]$ is a constant once computed.

$\therefore E[E[X]] = E[X]$

$(\because E[c] = c)$

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Find 2nd, 3rd moment and variance.

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$$= p$$

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$$VAR[X] = E[X^2] - E[X]^2$$

$$= p - p^2$$

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$$\frac{\partial}{\partial p}(p - p^2) = 0 \Rightarrow 1 - 2p = 0$$

$$\Rightarrow p = 0.5$$

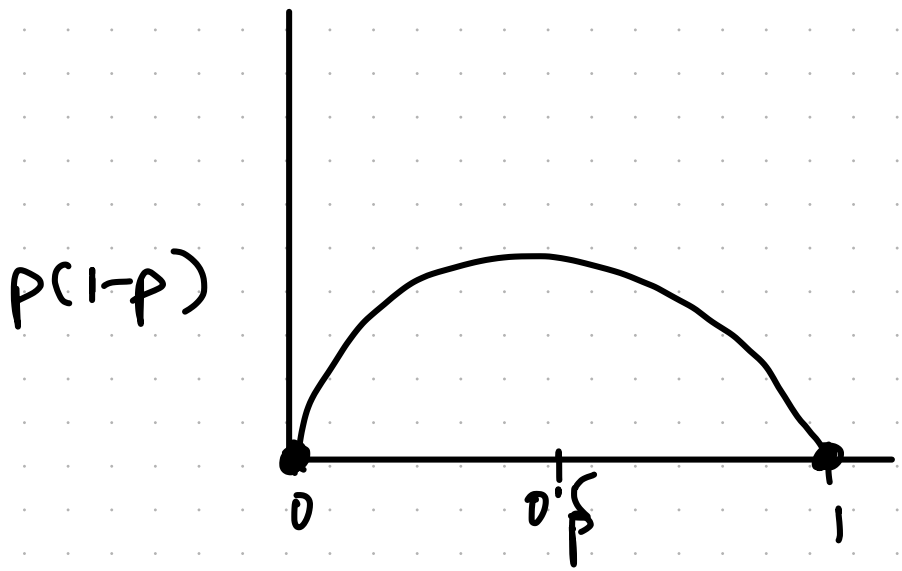
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$= \sum_{k=1}^{n} \frac{k(k-1)\, n!\, p^k (1-p)^{n-k}}{(n-k)!\, k(k-1)!} + \sum_{k=1}^{n} k \frac{n!}{(n-k)!\, k!} p^k (1-p)^{n-k}$

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$= \sum_{k=2}^{n} \frac{n!\, p^k (1-p)^{n-k}}{(n-k)!\,(k-2)!} + E[X]$

(Q) $X \sim \text{Binomial}(n, p)$

$E[X] = np$

$\text{VAR}[X] = ?$

$\text{VAR}[X] = E[X^2] - E[X]^2$

Now, $E[X^2] = \sum_{k=2}^{n} \frac{k(k-1)\, n!\, p^k (1-p)^{n-k}}{(n-k)!\, k(k-1)!} + \sum_{k=1}^{n} k \frac{n!}{(n-k)!\, k!} p^k (1-p)^{n-k}$

$= \sum_{k=2}^{n} \frac{n!\, p^k (1-p)^{n-k}}{(n-k)!\,(k-2)!} + E[X]$

$= \sum_{k=2}^{n} n(n-1) \frac{(n-2)!\, p^k (1-p)^{n-k}}{(n-k)!\,(k-2)!} + E[X]$

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$$= \sum_{k=2}^{n} n(n-1) \frac{(n-2)!\, p^{k-2} \cdot (1-p)^{n-k}}{(n-k)!\,(k-2)!} \cdot p^2 + E[X]$$

$$= p^2 n(n-1) \sum_{k=2}^{n} \frac{(n-2)!\, p^{k-2} (1-p)^{n-k}}{(n-k)!\,(k-2)!} + E[X]$$

let $l = k-2$

$k_{min} = 2$ $\qquad$ $k_{max} = n$

$l_{min} = 0$ $\qquad$ $l_{max} = n-2 = m$

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Now, $E[X^2] = p^2 n(n-1) \sum_{l=0}^{m} \frac{m!\, p^l (1-p)^{m-l}}{(m-l)!\, l!} + E[X]$

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$\text{VAR}(X) = E[X^2] - E[X]^2$

$= p^2 n(n-1) + np - (np)^2$

$= p^2 n^2 - p^2 n + pn - n^2 p^2$

$\text{VAR}(X) = np(1-p)$

Properties of Variance

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Properties of Variance

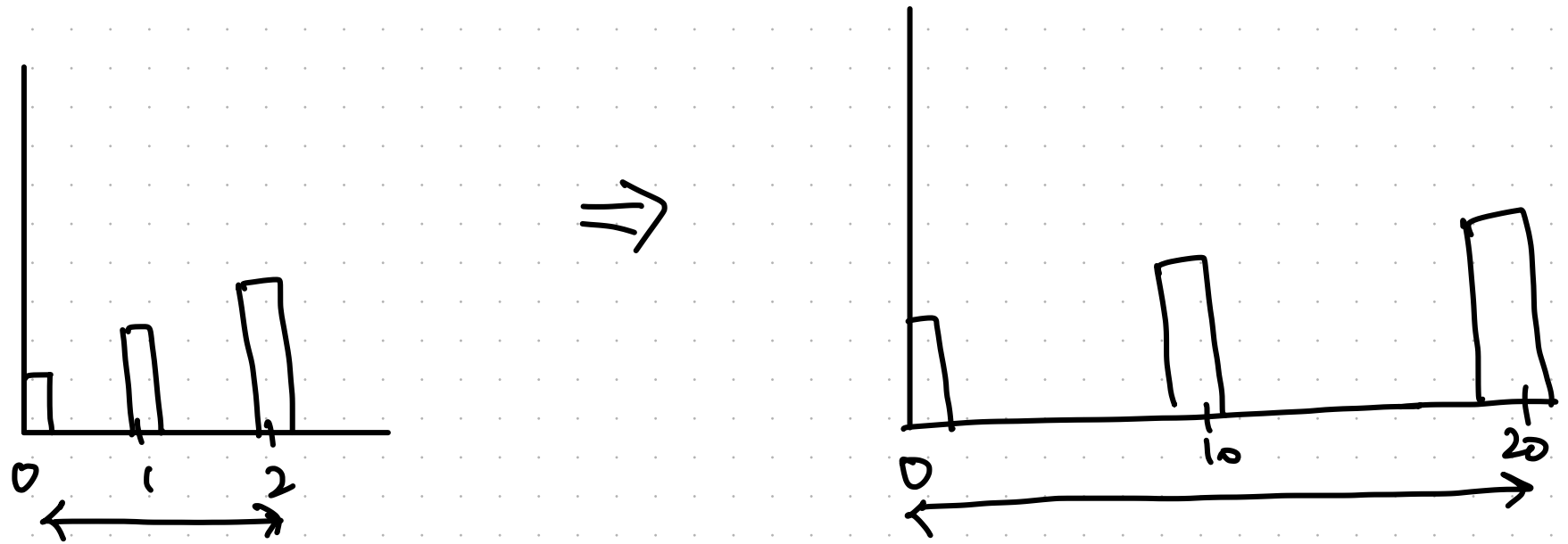
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$$= E[X^2] + c^2 + 2c\,E[X] - E[X]^2 - c^2 - 2c\,E[X]$$

$$= E[X^2] - E[X]^2$$

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